As we saw in Section 4.13, every solution to the time-independent Schrödinger equation allows us to construct to a time-dependent solution. All we need to do is multiply the time-independent solution—in this case $e^{\frac{ipx}{\hbar}}$—by $e^{-i\frac{Et}{\hbar}} = e^{-i\frac{p^2t}{2m\hbar}}$. Thus, a complete set of solutions can be written as

$$\psi(x,t) = \exp \frac{i(px - \frac{p^2t}{2m})}{\hbar}.$$

Any solution is a sum, or integral, of these solutions:

$$\psi(x,t) = \int \tilde{\psi}(p) \left(\exp \frac{i(px - \frac{p^2t}{2m})}{\hbar} \right) dp.$$

You can start with any wave function at $t = 0$, find $\tilde{\psi}(p)$ by Fourier transform, and let it evolve. The shape will change because the waves for different p values travel at different velocities. But, as we will soon see, the overall wave packet will travel at velocity $\langle p/m \rangle$ just as a classical particle would.

This simple general solution has an important implication. Among other things, it says that the momentum-representation wave function changes with time in a very simple way:

$$\tilde{\psi}(p,t) = \tilde{\psi}(p) \exp \frac{i(px - \frac{p^2t}{2m})}{\hbar}.$$

In other words, only the phase changes with time, while the magnitude remains constant. What makes this so interesting is that the probability $P(p)$ does not change at all with time. This, of course, is a consequence of momentum conservation, but it only holds if there are no forces acting on the particle.